

Digital Systems and Binary Numbers

INSTRUCTOR: MEHVISH FATIMA

Learning Objectives

- In this chapter you will learn about:
 - Non-positional number system
 - Positional number system
 - Decimal number system
 - Binary number system
 - Octal number system
 - Hexadecimal number system



Learning Objectives

- Convert a number's base
 - Another base to decimal base
 - Decimal base to another base
 - Some base to another base
- Shortcut methods for converting
 - Binary to octal number
 - Octal to binary number
 - Binary to hexadecimal number
 - Hexadecimal to binary number
- Fractional numbers in binary number system



Number Systems

•**Two types of number systems are:**

- Non-positional number systems
- Positional number systems

Non-positional Number Systems

- **Characteristics**

- Use symbols such as I for 1, II for 2, III for 3, IIII for 4, IIIII for 5, etc
- Each symbol represents the same value regardless of its position in the number
- The symbols are simply added to find out the value of a particular number

- **Difficulty**

- It is difficult to perform arithmetic with such a number system

Positional Number Systems

- **Characteristics**
 - Use only a few symbols called digits
 - These symbols represent different values depending on the position they occupy in the number

Positional Number Systems

- The value of each digit is determined by:
 1. The digit itself
 2. The position of the digit in the number
 3. The base of the number system

(**base** = total number of digits in the number system)

- The maximum value of a single digit is always equal to one less than the value of the base

Decimal Number System

Characteristics

- A positional number system
- Has 10 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9). Hence, its base = 10
- The maximum value of a single digit is 9 (one less than the value of the base)
- Each position of a digit represents a specific power of the base(10)
- We use this number system in our day-to-day life

Decimal Number System

Example

$$2586_{10} = (2 \times 10^3) + (5 \times 10^2) + (8 \times 10^1) + (6 \times 10^0)$$

$$= 2000 + 500 + 80 + 6$$

Binary Number System

- **Characteristics**

- A positional number system
- Has only 2 symbols or digits (0 and 1). Hence its base = 2
- The maximum value of a single digit is 1 (one less than the value of the base)
- Each position of a digit represents a specific power of the base (2)
- This number system is used in computers

Binary Number System

Example

$$\begin{aligned} 10101_2 &= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= 16 + 0 + 4 + 0 + 1 \\ &= 21_{10} \end{aligned}$$

Representing Numbers in Different Number Systems

In order to be specific about which number system we are referring to, it is a common practice to indicate the base as a subscript. Thus, we write:

$$10101_2 = 21_{10}$$

Bit

- Bit stands for **binary** digit
- A bit in computer terminology means either a 0 or a 1
- A binary number consisting of n bits is called an n -bit number

Octal Number System

Characteristics

- A positional number system
- Has total 8 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7). Hence, its base = 8
- The maximum value of a single digit is 7 (one less than the value of the base)
- Each position of a digit represents a specific power of the base (8)

Octal Number System

(Continued from previous slide..)

- Since there are only 8 digits, 3 bits ($2^3 = 8$) are sufficient to represent any octal number in binary

Example

$$\begin{aligned} 2057_8 &= (2 \times 8^3) + (0 \times 8^2) + (5 \times 8^1) + (7 \times 8^0) \\ &= 1024 + 0 + 40 + 7 \\ &= 1071_{10} \end{aligned}$$

Hexadecimal Number System

Characteristics

- A positional number system
- Has total 16 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F). Hence its base = 16
- The symbols A, B, C, D, E and F represent the decimal values 10, 11, 12, 13, 14 and 15 respectively
- The maximum value of a single digit is 15 (one less than the value of the base)

Hexadecimal Number System

- Each position of a digit represents a specific power of the base (16)
- Since there are only 16 digits, 4 bits ($2^4 = 16$) are sufficient to represent any hexadecimal number in binary

Example

$$\begin{aligned} 1AF_{16} &= (1 \times 16^2) + (A \times 16^1) + (F \times 16^0) \\ &= 1 \times 256 + 10 \times 16 + 15 \times 1 \\ &= 256 + 160 + 15 \\ &= 431_{10} \end{aligned}$$

Positional No:

1. Decimal No (Base =10, 0-9)
2. Binary (Base=2, 0 and 1)
3. Octal (Base=8, 0,1,2,3,4,5,6,7)
4. Hexadecimal (Base=16, 0-9,A,B,C,D,E and F)

Converting a Number of Another Base to a Decimal Number

Method

- Step 1: Determine the column (positional) value of each digit
- Step 2: Multiply the obtained column values by the digits in the corresponding columns
- Step 3: Calculate the sum of these products

Converting a Number of Another Base to a Decimal Number

(Continued from previous slide..)

Example

$$4706_8 = ?_{10}$$

Common

$$\begin{aligned} 4706_8 &= \overset{\substack{\downarrow \text{values} \\ 4}}{4} \times \overset{\substack{\downarrow \\ 8^3}}{8^3} + \overset{\substack{\downarrow \\ 7}}{7} \times \overset{\substack{\downarrow \\ 8^2}}{8^2} + \overset{\substack{\downarrow \\ 0}}{0} \times \overset{\substack{\downarrow \\ 8^1}}{8^1} + \overset{\substack{\downarrow \\ 6}}{6} \times 8^0 \end{aligned}$$

multiplied by the corresponding digits

$$\begin{aligned} &= 4 \times 512 + 7 \times 64 + 0 + 6 \times 1 \\ &= 2048 + 448 + \overset{\longleftarrow}{0} + 6 \end{aligned}$$

Sum of these products

$$= 2502_{10}$$

Converting a Decimal Number to a Number of Another Base

Division-Remainder Method

Step 1: Divide the decimal number to be converted by the value of the new base

Step 2: Record the remainder from Step 1 as the rightmost digit (least significant digit) of the new base number

Step 3: Divide the quotient of the previous divide by the new base

Converting a Decimal Number to a Number of Another Base

Step 4: Record the remainder from Step 3 as the next digit (to the left) of the new base number

Repeat Steps 3 and 4, recording remainders from right to left, until the quotient becomes zero in Step 3

Note that the last remainder thus obtained will be the most significant digit (MSD) of the new base number

Converting a Decimal Number to a Number of Another Base

(Continued from previous slide..)

Example

$$952_{10} = ?_8$$

Solution:

8	952	Remainder
	119	0
	14	7
	1	6
	0	1

$$\text{Hence, } 952_{10} = 1670_8$$

Converting a Number of Some Base to a Number of Another Base

Method

Step 1: Convert the original number to a decimal number (base 10)

Step 2: Convert the decimal number so obtained to the new base number

Converting a Number of Some Base to a Number of Another Base

Example

$$545_6 = ?_4$$

Solution:

Step 1: Convert from base 6 to base 10

$$\begin{aligned} 545_6 &= 5 \times 6^2 + 4 \times 6^1 + 5 \times 6^0 \\ &= 5 \times 36 + 4 \times 6 + 5 \times 1 \\ &= 180 + 24 + 5 \\ &= 209_{10} \end{aligned}$$

Converting a Number of Some Base to a Number of Another Base

Step 2: Convert 209_{10} to base 4

4	209	Remainders
	52	1
	13	0
	3	1
	0	3

Hence, $209_{10} = 3101_4$

So, $545_6 = 209_{10} = 3101_4$

Thus, $545_6 = 3101_4$

Shortcut Method for Converting a Binary Number to its Equivalent Octal Number

Method

Step 1: Divide the digits into groups of three starting from the right

Step 2: Convert each group of three binary digits to one octal digit using the method of binary to decimal conversion

Shortcut Method for Converting a Binary Number to its Equivalent Octal Number

Example

$$1101010_2 = ?_8$$

Step 1: Divide the binary digits into groups of 3 starting from right

001 101010

Step 2: Convert each group into one octal digit

$$001_2 = 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1$$

$$101_2 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 5$$

$$010_2 = 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2$$

$$\text{Hence, } 1101010_2 = 152_8$$

Shortcut Method for Converting an Octal Number to Its Equivalent Binary Number

Method

Step 1: Convert each octal digit to a 3 digit binary number (the octal digits may be treated as decimal for this conversion)

Step 2: Combine all the binary groups
resulting
(of 3 each) into a single binary
digits
number

Shortcut Method for Converting an Octal Number to Its Equivalent Binary Number

Example

$$562_8 = ?_2$$

Step 1: Convert each octal digit to 3 binary digits $5_8 = 101_2$, $6_8 = 110_2$, $2_8 = 010_2$

Step 2: Combine the binary groups $562_8 =$
 $\underline{101} \quad \underline{110} \quad \underline{010}$
 5 6 2

$$\text{Hence, } 562_8 = 101110010_2$$

Shortcut Method for Converting a Binary Number to its Equivalent Hexadecimal Number

Method

- Step 1: Divide the binary digits into groups of four starting from the right
- Step 2: Combine each group of four binary digits to one hexadecimal digit

Shortcut Method for Converting a Binary Number to its Equivalent Hexadecimal Number

(Continued from previous slide..)

Example

$$111101_2 = ?_{16}$$

Step 1: Divide the binary digits into groups of four starting from the right

0011 1101

Step 2: Convert each group into a hexadecimal digit

$$\begin{aligned} 0011_2 &= 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 3_{10} = 3_{16} \\ 1101_2 &= 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13_{10} = D_{16} \end{aligned}$$

$$\text{Hence, } 111101_2 = 3D_{16}$$

Shortcut Method for Converting a Hexadecimal Number to its Equivalent Binary Number

Method

Step 1: Convert the decimal equivalent of each hexadecimal digit to a 4 digit binary number

Step 2: Combine all the resulting binary groups

(of 4 digits each) in a single binary number

Shortcut Method for Converting a Hexadecimal Number to its Equivalent Binary Number

(Continued from previous slide..)

Example

$$2AB_{16} = ?_2$$

Step 1: Convert each hexadecimal digit to a 4 digit binary number

$$\begin{aligned} 2_{16} &= 2_{10} = 0010_2 \\ A_{16} &= 10_{10} = 1010_2 \\ B_{16} &= 11_{10} = 1011_2 \end{aligned}$$

Shortcut Method for Converting a Hexadecimal Number to its Equivalent Binary Number

(Continued from previous slide..)

Step 2: Combine the binary groups

$$2AB_{16} = \underbrace{0010}_2 \quad \underbrace{1010}_A \underbrace{1011}_B$$

$$\text{Hence, } 2AB_{16} = 001010101011_2$$

Fractional Numbers

Fractional numbers are formed same way as decimal number system

In general, a number in a number system with base b would be written as:

$$a_n a_{n-1} \dots a_0 . a_{-1} a_{-2} \dots a_{-m}$$

And would be interpreted to mean:

$$a_n \times b^n + a_{n-1} \times b^{n-1} + \dots + a_0 \times b^0 + a_{-1} \times b^{-1} + a_{-2} \times b^{-2} + \dots + a_{-m} \times b^{-m}$$

The symbols $a_n, a_{n-1}, \dots, a_{-m}$ in above representation should be one of the b symbols allowed in the number system

Formation of Fractional Numbers in Binary Number System (Example)

	Binary Point ↓									
Position	4	3	2	1	0	.	-1	-2	-3	-4
Position Value	2^4	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}	2^{-4}
Quantity Represented	16	8	4	2	1		$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$

Representation of Fractional Numbers in Binary Number System (Example)

(Continued from previous slide..)

Example

$$110.101_2$$

$$= 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 4 + 2 + 0 + 0.5 + 0 + 0.125$$

$$= 6.625_{10}$$

Formation of Fractional Numbers in Octal Number System (Example)

	Octal Point ↓							
Position	3	2	1	0	.	-1	-2	-3
Position Value	8^3	8^2	8^1	8^0		8^{-1}	8^{-2}	8^{-3}
Quantity Represented	51264	8	1			$1/8$	$1/64$	$1/512$

Formation of Fractional Numbers in Octal Number System (Example)

Example

$$\begin{aligned} 127.54_8 &= 1 \times 8^2 + 2 \times 8^1 + 7 \times 8^0 + 5 \times 8^{-1} + 4 \times 8^{-2} \\ &= 64 + 16 + 7 + \frac{5}{8} + \frac{4}{64} \\ &= 87 + 0.625 + 0.0625 \\ &= 87.6875_{10} \end{aligned}$$

End Lecture

